

Chapter 1

Lectures 6-10

1.1 February 2, 2026

Bellman Optimality Equation

$$V^*(s) = \max_a \left(r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right)$$

The statement that this is a fixed-point has nothing to do with contraction. We want to perform an iterative loop using the Bellman operator. The optimal value function V^* is the unique fixed point of the Bellman operator T , i.e., $TV^* = V^*$. Value iteration, defined as $V_{k+1} = TV_k$, with an arbitrary initial value V_0 will converge to the optimal value function V^* , i.e. $\lim_{k \rightarrow \infty} V_k = V^*$.

Optimal policy π^* can be computed as

$$\pi^*(s) = \arg \max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^*(s')])$$

Optimal Q-value function of a policy π

$$Q_\pi(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = 0 \right]$$

Just like optimal value function, you can define an optimal Q-value function

$$Q^*(s, a) = \max_{\pi} Q_\pi(s, a)$$

There exists a conversion from V^* to Q^* through

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^*(s')]$$

And from Q^* to V^*

$$V^*(s) = \max_a Q^*(s, a)$$

Also from π^* to Q^*

$$\pi^*(s) = \arg \max_a Q^*(s, a)$$

In some problems, you may not know the transition probability.

Optimal Q-value function Q^* satisfies the Bellman equation

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [\max_b Q^*(s', b)]$$

Define the mapping $F : \mathbb{R}^{|S||A|} \rightarrow \mathbb{R}^{|S||A|}$ as

$$(FQ)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [\max_b Q(s', b)]$$

1.1.1 Policy Iteration

Recap of policy iteration: at every point, we evaluate how good our situation is (policy, π_k). Our value function V_{π} satisfies the equation $V_{\pi} = R_{\pi} + \gamma P_{\pi} V_{\pi}$. How do we find V_{π} ?

Can we find the optimal policy by 'cleverly' searching over the policy space?

Policy iteration: for $k = 0, 1, 2, \dots$ Policy evaluation: compute the value of the policy to get V_{π_k}

$$\text{Solve } T_{\pi_k} V_{\pi_k} = V_{\pi_k}$$

Policy improvement: perform greedy update to get the next policy π_{k+1} :

$$\pi_{k+1}(x) = \arg \max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_{\pi_k}(s')])$$

Proposition (convergence of policy iteration): $V_{\pi_{k+a}} \geq V_{\pi_k}$ and $\|V_{\pi_{k+1}} - V^*\|_{\infty} \leq \gamma \|V_{\pi_k} - V^*\|_{\infty}$.

Policy iteration: Lemma (monotone property of T_{π}): let $V_1 \geq V_2$. Then, $T_{\pi} V_1 \geq T_{\pi} V_2$.

Proof: note that $V_{\pi_k} = T_{\pi_k} V_{\pi_k} \leq T V_{\pi_k} = T_{\pi_{k+1}} V_{\pi_k}$

Using the monotone property of T_{π} ,

$$V_{\pi_k} \leq T_{\pi_{k+1}} V_{\pi_k} \leq T_{\pi_{k+1}}^{(1)} V_{\pi_k} \leq \dots \leq T_{\pi_{k+1}}^{(2)} \dots$$